Evan Hobson

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Education

Brigham Young University, Provo, Utah — 2021 – Present
B.S. Wildlife & Wildlands Conservation — *Expected Graduation: April 2027*

Focus: Quantitative ecology, statistical and spatial modeling

Selected Coursework: Geographic Information Systems; Wildlife Behavioral Ecology; Scientific Computing; Linear Algebra; Computational Linear Algebra; Data Structures; Introductory Computer Programming; Principles of Wildlife & Fishery Management

Research Interests

Ecological modeling and statistical inference; spatial modeling; how model specification and data-product choices change ecological conclusions; remote sensing of rangeland vegetation; habitat selection (RSF/SSF/iSSF); applied management on working landscapes.

Research Experience

BYU Rangeland Ecology Lab — Provo, UT

Research Assistant — Sep 2025 – Present

- Lead analyst for a GPS-collar study of cattle movement and grazing behavior using resource- and integrated step-selection functions (RSF/iSSF)
- Designed and maintained reproducible spatial modeling pipelines (covariate generation, mixed-effects model fitting, cross-validation, spatial prediction) in **R** and **ArcGIS**
- Analyzed GPS and accelerometer-linked telemetry from **64 cow-year deployments** (31 cows in 2021, 33 in 2022) on a 998-ha sagebrush steppe allotment in southeastern Idaho (5-min fixes in 2021; 7-min fixes in 2022)
- Compared three remote-sensing annual-grass products within identical RSF structures to test whether inferred cattle response to cheatgrass depends on product choice
- Produced management-relevant selection maps linking space use to topography, water access, annual grass (cheatgrass) cover, perennial forage, and weather covariates to inform targeted grazing strategies
- Conducted vegetation monitoring across two large rangeland field projects (summer 2026): the long-term Sagebrush Steppe Treatment Evaluation Project (SageSTEP) and a targeted sheep-grazing study in Idaho
- Measured plant density, biomass, and shrub volume; canopy and basal gap; foliar cover by line-point intercept (BLM AIM); and soil-stability/erosion tests across grass, forb, and shrub functional groups
- Awarded **BYU College Undergraduate Research Award (CURA; $4,000)** supporting analysis, manuscript preparation, and required conference presentation
- Re-established and advanced analysis of legacy feral horse telemetry datasets for manuscript development

BYU Wildlife Ecology Lab — Provo, UT

Research Assistant — Jul 2024 – Jul 2025

- Conducted field research on pygmy rabbit (*Brachylagus idahoensis*) ecology, including camera deployment, vegetation sampling, and habitat covariate collection
- Assisted with mule deer GPS collar retrievals, mortality investigations, and tissue collection
- Performed cementum annuli aging and microscopy-based laboratory workflows

Selected Research Projects

Cattle Resource Selection in Cheatgrass-Invaded Rangelands **2025 – Present**

- Lead analyst evaluating cattle use of cheatgrass during dormant-season targeted grazing on sagebrush steppe in southeastern Idaho
- Built three canonical RSFs with identical structure, substituting only the cheatgrass covariate, to test whether inference depends on remote-sensing product (RAP 10-m invasive annual grass, RAP 30-m annual forb-and-grass, USGS REPMS *Bromus tectorum*)
- Fit binomial GLMMs (glmmTMB) with a per-cow-year random cheatgrass slope and year interaction; modeled elevation, slope, northness, distance to water, and perennial forage biomass; validated with cow-grouped cross-validation and the Boyce index
- Manuscript with coauthors for review (lead author); target *Rangeland Ecology & Management*

Accelerometer Behavioral States and Habitat Selection **2025 – on hold**

- Testing whether cattle habitat selection differs by accelerometer-derived behavioral state (grazing, resting, walking)
- On hold pending the source study's original behavior-label records

Feral Horse Movement and Space Use in Northwestern Nevada **2025 – paused**

- Analyzed legacy GPS telemetry from 11 feral horses (30-min fixes; 2 months–2 years)
- Quantified daily distance traveled and daily space use using averaged daily minimum convex polygons (MCPs)
- Employed MCPs as a non-traditional, derived movement metric rather than for home-range inference
- Interpreted movement patterns relative to water availability and seasonal aridity
- Manuscript paused (lead author)

Research Outputs

Manuscripts (Lead Author)

- Hobson, E., et al. *Cattle resource selection in cheatgrass-invaded rangelands during dormant-season grazing*. With coauthors for review; target *Rangeland Ecology & Management*.
- Hobson, E., et al. *Seasonal and monthly variation in feral horse movement and space use in northwestern Nevada*. Paused.

Other Scholarly Writing & Presentations

Hobson, E. *Permission to Feel, Power to Become: How emotional engagement can enable spiritual growth.*
BYU Student Religion Symposium; accepted for presentation and invited for submission to symposium proceedings (non–peer reviewed).

Additional Experience

Western North American Naturalist **Provo, UT**

Assistant to Editor-in-Chief Apr 2025 – Jan 2026

- Managed dispersal of **$10,000+** in grant funds supporting journal operations
- Coordinated **50+ peer reviewers**, improving review turnaround and editorial workflow efficiency
- Gained experience with peer review, editorial decision-making, and scientific publishing standards

BYU Mathematics I.T. Department **Provo, UT**

Lead Student Computer Specialist May 2023 – Dec 2023

- Led technical support for faculty and applied mathematics labs
- Supported Linux system administration and computational lab transitions

Technical Skills

Modeling & Inference: RSF, SSF/iSSF, GLMs and mixed-effects models (glmmTMB), conditional logistic regression, use–availability designs, cluster-robust and random-effects inference, cross-validation, Boyce index validation, log-RSS, spatial prediction

GIS & Spatial Data: ArcGIS; raster/vector workflows; covariate derivation; cartographic outputs

Programming: R (primary); Python; Bash; Git; exposure to C++ and Julia through coursework

Telemetry & Field: GPS collars; accelerometers and behavioral-state classification; camera trapping; rangeland vegetation monitoring: line-point intercept (BLM AIM), canopy/basal gap, density, biomass, and shrub volume, soil-stability/erosion tests, plant functional-group ID

Scientific Writing: Manuscript drafting; figures; peer-review workflows